

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. For what positive numbers a does $\int_1^{\infty} a^x dx$ converge?

2. Consider each of the following questions.

a) Show that $\int_{-\infty}^{\infty} x dx$ is divergent.

b) Show that $\lim_{t \rightarrow \infty} \left[\int_{-t}^t x dx \right] = 0$

c) What do parts (a) and (b) show us?

3. If $\int_{-\infty}^{\infty} f(x) dx$ is convergent, a and b are real numbers, and $f(x)$ is continuous with an antiderivative of $F(x)$, show that $\int_{-\infty}^a f(x) dx + \int_a^{\infty} f(x) dx = \int_{-\infty}^b f(x) dx + \int_b^{\infty} f(x) dx$ regardless of the values of a and b .

4. An important concept in Engineering is the *Laplace transform* of f , which is the function F defined by $s > 0$ by

$$F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

- a) Find the Laplace transform (that is find a simplified form of $F(s)$) when $f(t) = 1$.

b) Find the Laplace transform (that is find a simplified form of $F(s)$) when $f(t) = t$.

c) Find the Laplace transform (that is find a simplified form of $F(s)$) when $f(t) = e^{at}$ (where a is just some real number).

5. Show how the Comparison Theorem can be used to help determine if the following integrals converge or diverge.

a) $\int_1^{\infty} \frac{1}{[x + \ln(x)]^2} dx$

b) $\int_1^{\infty} \frac{x+1}{\sqrt{x^4-x}} dx$

c) $\int_0^{\infty} \frac{x}{x^4+1} dx$

d) $\int_0^1 \frac{\sec^2(x)}{x\sqrt{x}} dx$

e) $\int_3^{\infty} \frac{x^3 - x + 2}{x^5 + x^4 - x^3} dx$

f) $\int_3^{\infty} \frac{1}{\cos(x) - 2 + \ln(x)} dx$

6. Suppose Kyle is trying to make the following comparison

$$\frac{x^3 - x + 2}{x^5 + x^4 - x^3} \leq \frac{x^3}{x^5 + x^4 - x^3} \leq \frac{x^3}{-x^3} = -1 \text{ where } x \text{ represents values on the interval } [2, \infty).$$

Is this comparison valid? Why or why not?